

Reviewer Count Is Not Evidence Count: Measured Error Dependence in an Executable Verification Panel

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September 20, 2026

Abstract

Panels of automated verifiers — test oracles, static analysers, and increasingly LLM reviewers — are routinely sized by head count, on the implicit assumption that adding a verifier adds evidence. We measure that assumption on a panel whose ground truth is decided by execution rather than by judgement: 25 partial test oracles voting on 72 program candidates drawn from 24 problems, where a candidate’s viability is the exit status of a hidden test suite. Every verifier discriminates significantly above chance (mean accuracy $p = 0.7956$, 25/25 significant Youden’s J after Bonferroni correction), so the panel’s dependence structure is interpretable rather than an artifact of incompetent members.

Three results follow. First, the panel buys nothing over its own members: majority-vote error is 0.2083 (95% CI [0.131, 0.316]) against a mean member’s 0.2044 and a *best* member’s 0.1389 — so the panel is not worth one verifier, it is worse than one, and its effective independent size falls below the metric’s floor of 1 rather than landing on it. On 72 items that ordering is directional, not significant (panel vs. best member, exact McNemar $p = 0.18$), and we report it as such. The design-effect heuristic $N/(1 + (N - 1)\rho)$, widely reused to discount correlated voters, reports 1.66 for this panel; under the dependence shape our data support it overstates available independence in 441/441 cells of a 63,000-run simulation grid, though a contemporaneous LLM-judge panel finds the same heuristic accurate at a lower correlation, so the failure is regime-specific rather than universal. Second, that shape is not the shared-shock mixture that recent multi-agent work assumes. Real panels fail *partially* — 11 of the 15 observed panel failures had a majority wrong while a minority was still right — an outcome shared shock assigns essentially zero probability. A per-item difficulty model in the family proposed by Eckhardt and Lee in 1985, at identical parameter count, reproduces the partial failures (11.1 predicted against 11 observed) and lands $3.7\times$ closer on panel error. Third, neither two-parameter model gets the high-quorum limit right, because the real panel has an *irreducible blind spot*: four of 72 candidates are missed by all 25 verifiers, a floor of $\lambda = 0.0556$ (95% CI [0.022, 0.134]) that no acceptance threshold and no panel size can cross. Shared shock over-predicts that floor by $2.13\times$ and per-item difficulty promises it away entirely.

The practical consequence is that a verification panel must be characterised by measurement, not by declared diversity: verifiers sharing no declared attribute in our panel still shared 53% of their errors, and weighting votes by declared independence groups — an intuitively safe defence — never wins when that declared structure is not real. Every finding rests on one constructed panel of 25 oracles over 72 items; all code, artifacts, and the analysis pipeline are public.

1 Introduction

Software increasingly arrives faster than people can review it. The response, across code review, CI, benchmark scoring, and agentic coding pipelines, has been to add reviewers: more static analysers, more test oracles, more model instances voting on the same artifact, aggregated by majority or by

quorum. The reasoning is the Condorcet jury theorem’s: individually fallible voters, combined, are reliably right.

That theorem’s premise is independence, which both its classical statement and its generalisations call unrealistic — Condorcet’s assumptions of independence and homogeneity are “seldom realistic” [3], and the theorem “is known to hold only under the unrealistic assumption that votes are independent” [12]. Both papers relax that premise rather than abandon it, deriving conditions under which majority voting survives dependence. Software engineering learned this the hard way once already. N-version programming [1] runs multiple independently developed implementations of one specification and votes on their outputs; in the words of the experiment that tested it, the method “depends for its reliability improvement on the assumption that programs that have been developed independently will fail independently” [9]. Knight and Leveson [9] then tested exactly that with 27 versions written by different people from the same specification, and found the coincident failure rate far above what independence predicts. The theoretical accounts that followed [5, 13] replaced the independence assumption with a per-input difficulty function: some inputs are simply hard, and every version tends to fail on them together.

The panels being assembled today are not independently written programs by different teams, but they inherit the same question and answer it less carefully. A modern verification panel is often five prompts against one model family, or five analysers sharing a parser, or five test oracles sampling one input domain. Its head count is reported; its independence is assumed. Where a discount is applied at all, it is the survey-sampling design effect $n_{\text{eff}} = N/(1 + (N - 1)\rho)$ [8], carried over from cluster sampling and evaluated at some estimate of inter-rater agreement — Kohli [10] apply it to an LLM-judge panel in exactly this way.

This paper asks the quantitative version of the question on a panel where the answer can be checked:

Given n verifiers with per-verifier accuracy p and measured error correlation ρ , how many statistically independent verifiers is that panel actually worth — and does any standard model of verifier dependence predict what a real panel does?

Contributions.

1. **An instrument where dependence is measurable** (§3). A 25-verifier panel of partial test oracles over 72 candidates with execution-decided ground truth. We first report the negative result that motivated it: a 20-member panel of four small open-weight LLMs across five prompt templates produced no member that discriminated above chance, making its correlation structure uninterpretable.
2. **A panel that loses to its own members** (§4). Majority vote errs on 0.2083 of items against the best member’s 0.1389, so the panel’s effective independent size falls *below* the metric’s floor of 1. Declared independence labels carry real but partial signal ($\rho = 0.892$ within group against 0.526 across). Kohli [10] reported the same headline on an LLM-judge panel while this work was in preparation, and has priority on that framing; §4.4 reconciles the two, including where they disagree.
3. **A model-shape result** (§5). The shared-shock mixture is not merely mis-parameterised but structurally wrong: it cannot produce partial panel failure, which is the dominant real failure mode. A per-item difficulty (beta-binomial) model at equal parameter count fits.
4. **An irreducible blind spot** (§6). The real quorum frontier bottoms out at $\lambda = 0.0556$ and stops. Neither two-parameter model expresses this; a one-inflated extension does. Escaping a correlation and escaping a blind spot require different interventions.
5. **Design consequences** (§7): a negative result on group-balanced aggregation, and a single-factor decomposition showing that panel *correlation* — not accuracy, and not the blind spot — carries 53 of the 58 catastrophic outcomes an optimising adversary induces.

Everything reported here is reproducible from committed artifacts in the IDKMesh repository [16]; §9 gives the commands.

2 Background and related work

Correlated failure in redundant software. The closest prior work is the multiversion programming literature. Knight and Leveson [9] is the empirical anchor: independence, assumed as a design property, did not hold. Eckhardt and Lee [5] give the model we find fits — inputs carry a difficulty $\theta(x)$, versions fail independently *conditional* on x , and the resulting marginal failures are positively correlated. Littlewood and Miller [13] extend it to versions built by deliberately different methodologies.

Two of Eckhardt and Lee [5]’s consequences deserve to be stated here, because this paper’s headline results are empirical instances of both. First, they show (§V, Fig. 5) that *certain intensity distributions make an N -version system, on average, more prone to failure than a single component* — redundancy reducing reliability. §4 reports a measured panel that does exactly this. Second, they show that “any error region for which a majority of software components fail will limit the effectiveness of redundancy regardless of the magnitude of the average component failure probability”, with the limiting failure probability governed only by intensity mass at $\theta \geq 0.5$ (their Eq. 15). §6’s irreducible floor λ is a measured instance of that region.

Our contribution here is therefore not the model, and not the phenomena — both were derived in 1985. It is that this model, proposed for human-written program versions and rarely tested since, is the one that fits an automated verification panel; that the alternative used in recent multi-agent and ensemble work does not; and that the two predicted pathologies are observable on a real panel of programs. We use “difficulty” where Eckhardt and Lee write “intensity”; the quantity is the same.

Aggregation and the independence discount. The Condorcet jury theorem under correlated votes has a substantial literature [3, 12]; the discount applied in practice is Kish’s design effect [8], imported from cluster sampling and used on a judge panel by Kohli [10]. Beta-binomial models for over-dispersed binary data are standard [17]; agreement statistics such as Fleiss’ κ [6] summarise rater concordance but do not, by themselves, answer how much independent evidence a panel holds. Ensemble learning inherited this model rather than reinventing it, and the chain matters because it ends at the diagnostic we use. Hansen and Salamon [7] carried Eckhardt and Lee [5] into classifier ensembles explicitly — “Eckhardt and Lee have proposed a model incorporating such input specific difficulty within a study of N -version programming” — and named the distribution over θ the *classification difficulty distribution*, which is Eckhardt and Lee’s intensity distribution G under another name. Kuncheva and Whitaker [11] then reduce that distribution to a scalar diversity index, its variance, alongside nine others. Closest of those to our setting is Partridge and Krzanowski [15]’s *coincident-failure diversity*, which is defined on software versions and built directly on the failure-count distribution — the same histogram as our Figure 2. Their $\text{CFD} = \sum_{n=1}^N \frac{N-n}{N-1} f_n$ weights a failure by how *few* versions share it: it is 1 when every failure is unique to one version and 0 when all N fail together. Our §6 floor is the mass they assign zero.

The distribution-of-errors diagnostic of §5 is therefore well-established. What this literature does with it is score a set on a scalar; what we do is fit competing generative models to it and ask which reproduces a measured panel. Kuncheva and Whitaker [11] also report a finding we echo: such measures track ensemble gain less reliably than intuition suggests — in their controlled simulations all ten relate similarly to majority-vote improvement, but they close with doubts about the measures’ usefulness for building ensembles on real problems.

Review panels in practice. In three large open-source projects, review *coverage* and *participation* were found to relate to post-release defect counts [14]; qualitative study of industrial review

finds its benefits extend well beyond defect-finding, to knowledge transfer and team awareness [2]. Neither measures reviewer *head count* — McIntosh et al.’s participation metrics are self-approval, review speed, and discussion. Head count is the cruder proxy practice has adopted, and what it is worth is the subject of this paper. Automated judging with language models [19] made evaluation cheap enough to run at scale, and execution-based benchmarks [4] supply exactly the kind of decidable ground truth this study needs. What this paper adds to §4.4’s contemporaneous measurement is a panel whose ground truth is decided by execution, and a test of which dependence *model* reproduces it.

3 The instrument

3.1 A negative result first: LLM panels we could not measure

Our first attempt deployed live LLM verifiers, because that is the panel practitioners are actually building. Twenty verifiers — four open-weight models (`qwen2.5-1.5b`, `llama3.2-1b`, `gemma2-2b`, `smollm2-1.7b`) crossed with five prompt templates — voted on all 72 candidates, 1440 votes total.

The experiment failed, informatively. Of the 20 verifiers, six emitted a single constant verdict across all 72 tasks and three more were $\geq 95\%$ one verdict. Accuracy hid this: on a corpus that is 36.1% viable, “always reject” scores 0.639. Youden’s $J = \text{sensitivity} + \text{specificity} - 1$ [18] does not hide it. Youden [18] states the relevant property directly: J “has the value zero whenever a diagnostic test gives the same proportion of positives for both diseased and control groups regardless of what that proportion is” — so a rule that ignores the artifact scores zero whichever way it is biased. Mean J was $+0.0487$, and **zero of 20 verifiers had J significantly above zero after Bonferroni correction**. The 20-member majority scored accuracy 0.514 against 0.639 for a constant rejector, with a permutation test on panel J giving $p = 0.13$.

What the votes *did* respond to was the prompt: mean accept rate moved from 0.251 to 0.731 across templates and 0.137 to 0.799 across models, while carrying no task-level signal. The panel measured its own prompts.

We report this because it constrains the paper’s scope and because it is a methodological warning: **a correlation estimated over verifiers that do not discriminate is a correlation between constants**. Any panel study must pass a discrimination screen before its dependence structure means anything, and accuracy on an unbalanced corpus is not that screen.

3.2 A panel of partial oracles

We therefore built verifiers that demonstrably verify. A verifier is a *partial test oracle*: it draws inputs from one named *region* of a problem’s input domain — `tiny`, `small`, `large`, `extreme`, `duplicate` — and accepts a candidate only if it matches a reference implementation on all drawn inputs. Five regions $\times$ five random seeds gives $n = 25$ verifiers.

The corpus is 24 problems $\times$ 3 variants = 72 candidate solutions. Ground truth is not a judgement: each candidate is executed against hidden tests in a subprocess and `viable` is the exit status. 26 are viable, 46 are not, a base rate of 0.639 non-viable. Labels reproduce exactly on regeneration (72/72).

Are the labels independent of the verifiers? They come from different artifacts. Ground truth is decided by executing *hand-written hidden tests* — literal input/expected pairs such as (`"solve([1,2,3,4])"`, `2.5`) — authored per problem from the specification. The verifiers never see those tests: a partial oracle compares the candidate’s output against the executed output of the problem’s *reference implementation* on inputs it draws itself. So the two sides are independent encodings of one specification, and neither is derived from the other; expected outputs are never hand-written for the oracles, and the hidden tests are never generated from the reference. Two further guards matter. The variant labels (`"ok"`, `"off_by_one"`) are *not* used as ground truth —

a mutant that happens to pass every hidden test is recorded viable — and inputs on which the reference itself raises are treated as out of contract rather than as defects. The residual shared dependency is the specification itself: a misconception present in both the reference and the hand-written expectations would corrupt both, and this design cannot detect that.

This panel has three properties that make it a usable instrument. Every verifier is a program, so every error is a genuine missed defect rather than a disagreement about taste. Every error vector is observed rather than assumed. And region membership is a *declared independence label* of exactly the kind panel designers rely on — “these reviewers look at different things” — which makes the label itself testable. §4 tests it and finds it wanting.

The panel passes the screen §3.1 failed: **25 of 25 verifiers discriminate above chance**, every Youden’s J significantly positive after Bonferroni correction, at mean accuracy $p = 0.7956$.

3.3 Effective independent panel size

For independent verifiers, the probability that a panel’s decision is wrong is a binomial tail. With $need = \lfloor qn \rfloor + 1$ votes required to accept at quorum q ,

$$E_{\text{indep}}(n, p, q) = \sum_{k=0}^{need-1} \binom{n}{k} p^k (1-p)^{n-k}, \quad (1)$$

which is strictly decreasing in n . We define the **effective independent panel size** of a measured correlated panel as the size an *independent* panel would need to reproduce the same measured error rate,

$$n_{\text{eff}}(n, p, \rho, q) = E_{\text{indep}}^{-1}(\text{measured error rate}), \quad (2)$$

interpolated linearly between bracketing odd panel sizes. The ratio n_{eff}/n is the panel’s **evidence efficiency**: the fraction of nominal reviewers that survives as independent evidence.

The metric saturates at both ends, and this matters. E_{indep} is evaluated over odd panel sizes $1, 3, \dots, 201$, so n_{eff} is *clamped*: it returns 1 for any measured error at or above the single-verifier error, and 201 for any error at or below the largest panel’s. The lower clamp is not cosmetic. A panel that performs *worse* than one of its members is not thereby “worth one member” — it is worth less, and the metric cannot say how much less. We therefore report n_{eff} only where it is unclamped, and elsewhere report the error rates directly. Of the 315 grid cells at $q = 0.5$, 28 (8.9%) sit on the lower clamp.

The metric is anchored where it is unclamped. On a simulated independent 5-panel at $p = 0.75, q = 0.5$ the analytic error is 0.103516 and the simulator measures 0.103521. Across the grid of §4.3 the $\rho = 0$ column recovers nominal panel size over a sevenfold range (2.98 to 20.99) without having been fitted (Figure 1). The $\rho = 1$ column reads 1.00 at every panel size, but that is the lower clamp firing rather than a measurement, and we do not claim it as validation.

4 How much independent evidence does the panel hold?

4.1 The declared label is real, and not nearly enough

Pairwise error correlations, over all $\binom{25}{2} = 300$ pairs:

pair type	pairs	mean ρ
same region (declared dependent)	50	+0.8924
different region (declared independent)	250	+0.5263
all pairs	300	+0.5873

The declared label carries genuine signal: +0.892 within a region against +0.526 across. But **verifiers that share no declared attribute still share 53% of their errors**. Treating a metadata boundary as an independence boundary would overstate the panel’s independent evidence by a wide margin. The practical form of this is a measurement instruction, not a policy: before weighting votes by declared groups, check whether cross-group error correlation is actually lower than within-group correlation, and by how much.

4.2 Headline: the panel is worse than its own members

	error on 72 items	95% CI
25-verifier majority	0.2083 (15/72)	[0.131, 0.316]
mean member	0.2044	—
best member (large-s0)	0.1389 (10/72)	[0.077, 0.237]
panel vs. best member: exact McNemar $p = 0.18$ ($b = 2, c = 7$)		
effective size n_{eff} : <i>below</i> the metric’s floor of 1 (clamped)		
design-effect heuristic $N/(1 + (N - 1)\rho)$: 1.66		

Under majority vote this panel does not merely fail to beat its members — it loses to them. Eckhardt and Lee [5] predicted that this is possible for sufficiently coincident intensity distributions (their Fig. 5, “an intensity distribution for which redundancy reduces reliability”); what follows is a measurement of it on a panel of real programs. Aggregating 25 verifiers produces *more* error than the mean member and half again the error of the best one. Because the panel error exceeds the single-verifier error, n_{eff} sits on its lower clamp (§3.3): the honest statement is that the panel’s effective independent size is below 1, not equal to it.

Two cautions belong with this number rather than in a later section. First, 72 items is a small sample: the panel–best-member gap is directional but not significant ($p = 0.18$), and we claim the *ordering*, not the magnitude. Second, the design-effect heuristic reports 1.66 where the measured value is below 1, so it overstates independence *here*; §4.3 shows when that generalises and §4.4 shows a measured panel where it does not.

4.3 When the heuristic is optimistic, and under what condition

To separate “this panel” from “panels”, we ran a full factorial simulation: $n \in \{3, 5, 7, 9, 11, 15, 21\}$, $p \in \{0.60, 0.70, 0.75, 0.80, 0.90\}$, $\rho \in \{0, 0.125, \dots, 1.0\}$, $q \in \{0.5, 0.7\}$, 100 seeds per cell — 630 cells, 63,000 runs. Figure 1 plots the resulting n_{eff} surface at $p = 0.75, q = 0.5$.

Two readings of this grid matter. The first is a ceiling: even 12.5% error correlation caps a 21-member panel at 10.6 effective verifiers, and at $\rho = 0.5, p = 0.75$ going from 15 to 21 reviewers buys +0.05 effective verifiers. The second concerns the heuristic. Re-deriving each cell under the two competing dependence models of §5 gives:

cells where $N/(1 + (N - 1)\rho)$ overstates independence	
under the shared-shock model	17/441 (4%)
under the per-item difficulty model	441/441 (100%)

Under the item-difficulty shape the design-effect heuristic overstates available independence in *every* cell. The rule this licenses is a conditional one, and the condition matters: **under item-difficulty dependence, $N/(1 + (N - 1)\rho)$ overstates a panel’s independent evidence — so establish the dependence shape before using it to size a panel**. We deliberately do not state this unconditionally. The 441/441 figure is computed *under* the item-difficulty model; comparing it against a heuristic derived under different assumptions demonstrates that two models disagree, not that the heuristic is false in general. §4.4 describes a measured panel where it is in fact accurate.

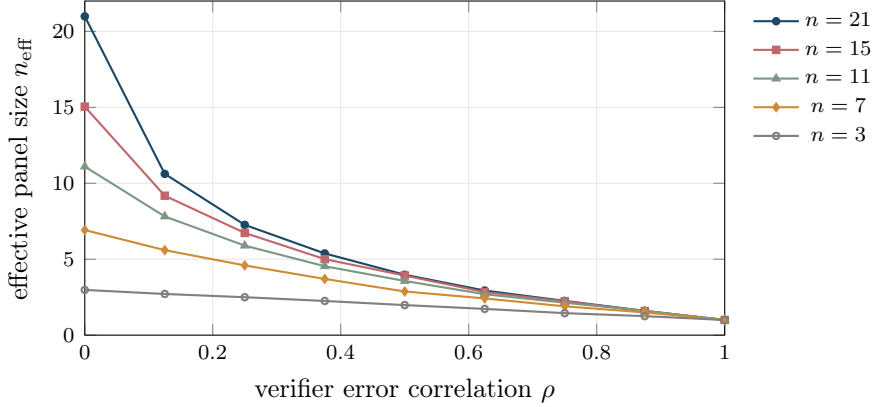


Figure 1: Effective independent panel size against error correlation, at $p = 0.75$, $q = 0.5$, from 63,000 simulation runs. At $\rho = 0$ the metric recovers nominal size across a sevenfold range (2.98 to 20.99) without having been fitted. The $\rho = 1$ column reads 1.00 everywhere, but that is the metric’s lower clamp (§3.3), not a measurement. The curves converge from $\rho \approx 0.5$ upward, where the spread across $n = 3 \dots 21$ falls below 2.0 effective verifiers; below that panel size still matters substantially (spread 7.9 at $\rho = 0.125$, 18.0 at $\rho = 0$).

This also corrects a conclusion we had drawn ourselves. Under shared shock, n_{eff} saturates as the panel grows and does so most sharply for accurate verifiers, suggesting that high accuracy is a reason to stop adding reviewers. Under per-item difficulty the regimes invert. Extending past the grid at $p = 0.90$, $\rho = 0.125$:

n	5	11	21	41	81	151
shared shock	3.82	4.57	4.59	4.59	4.59	4.59
per-item difficulty	2.82	3.93	4.56	4.89	5.16	5.40

Shared shock pins at 4.59 and never moves; per-item difficulty crosses it near $n = 21$ and keeps climbing. Conversely at $p = 0.75$, $\rho = 0.5$ the per-item model flattens almost immediately, at $n_{\text{eff}} \approx 1.6$ rather than the 4.1 shared shock reports. **High correlation is what caps a panel, and it caps it harder and sooner than the shared-shock model says; high accuracy is not itself a reason to stop adding verifiers.**

4.4 A contemporaneous panel where the heuristic holds

While this work was in preparation, Kohli [10] reported the same headline on a very different panel: nine frontier LLM judges from seven model families, on three natural-language-inference benchmarks with 100 human annotations per item. That panel is worth $n_{\text{eff}} = 2.18$ of 9 (bootstrap CI [2.07, 2.31]), the best single judge matches or beats the full panel, and the conclusion — that correlated judges, not the aggregation rule, are the binding constraint — is ours as well. Agreement between two independent instruments, one on code with executable ground truth and one on natural language with human labels, is the strongest evidence either paper offers that the effect is real rather than an artifact of one construction. That work has priority on the effective-votes framing for judge panels, and we make no first-measurement claim.

The two panels disagree on exactly two points, and the disagreement is informative.

The heuristic. Kohli [10] finds $N/(1 + (N - 1)\rho)$ *accurate* on their panel: the empirical $n_{\text{eff}}(k)$ curve tracks $k/(1 + (k - 1) \cdot 0.391)$, and an independent eigenvalue estimate (2.16) matches the Kish value (2.18). We find it optimistic on ours. The panels differ in measured correlation (0.391 against 0.5873) and, more consequentially, in dependence shape.

The shape. Their permutation test rejects shared item difficulty as the explanation of their inter-judge correlation ($p < 10^{-4}$; only 6.8–13.5% of the gap is attributable to it). We find item

difficulty is the shape that fits ours. The honest reading favours their result as a constraint on ours rather than the reverse, because *our panel’s item-difficulty structure is manufactured*: five input regions $\times$ five seeds induces per-item difficulty almost by construction, since a candidate that fails on **extreme** inputs fails for every verifier drawing from that region. Their panel’s diversity is not designed at all — it is whatever seven model families happen to share.

The defensible summary is therefore not that one paper is right, but that **dependence shape is a property of how a panel was assembled and must be measured per panel rather than assumed**. That is a weaker claim than either paper makes alone, and better supported than either.

5 The shape of verifier dependence

5.1 A single ρ does not reproduce the panel

Feeding the panel’s own measured correlation back into the shared-shock mixture — with probability ρ all verifiers share one correctness state, otherwise they are independent — under-predicts what the panel does:

model	panel error	vs. real
real 25-verifier majority	0.2083	—
independent ($\rho = 0$)	0.0005	417 $\times$ too low
flat shared shock at $\rho = 0.587$	0.1216	1.71 $\times$ too low
nested shared shock ($g = 0.526, b = 0.773$)	0.1290	1.62 $\times$ too low

A nested variant matching *both* the within-region and cross-region correlations barely improves on the flat one, and heterogeneous accuracy does not close the gap either: re-simulating with each verifier’s own measured accuracy gives 0.1228, still 0.086 short. The problem is not the parameter value.

5.2 Real panels fail partially; shared shock cannot

The diagnosis is visible in the distribution of how many verifiers err per task (Figure 2). That diagnostic is not new — it is the empirical form of Eckhardt and Lee [5]’s intensity distribution, called the classification difficulty distribution by Hansen and Salamon [7] and reduced to a scalar index by Kuncheva and Whitaker [11]. What has not been done, and is what follows, is to treat the distribution as something to be *explained*: to fit competing generative models to it at matched parameter count and ask which one reproduces a real panel.

Replacing “with probability ρ everyone shares one correctness state” with “each task has a difficulty $d \sim \text{Beta}(\alpha, \beta)$ and each verifier errs independently with probability d ” gives a beta-binomial [17]. This is Eckhardt and Lee [5]’s model with a Beta mixing distribution: their Corollary writes system failure as $p_N = \int h(y; N) dG(y)$, a binomial mixed over the *intensity distribution* G , and they deliberately leave G unspecified. Choosing $G = \text{Beta}$ is our parametric commitment, not theirs — a point §8 already concedes. It has the same two free parameters as flat shared shock — a mean and a correlation — so the comparison is of *shape*, not of freedom:

	real	per-item difficulty	shared shock
panel error at majority	0.2083	0.1847	0.1216
partial failures ($13 \leq k \leq 24$)	11	11.1	0.01
unanimous failures ($k = 25$)	4	2.3	8.5
implied pairwise ρ	0.5873	0.5713	—

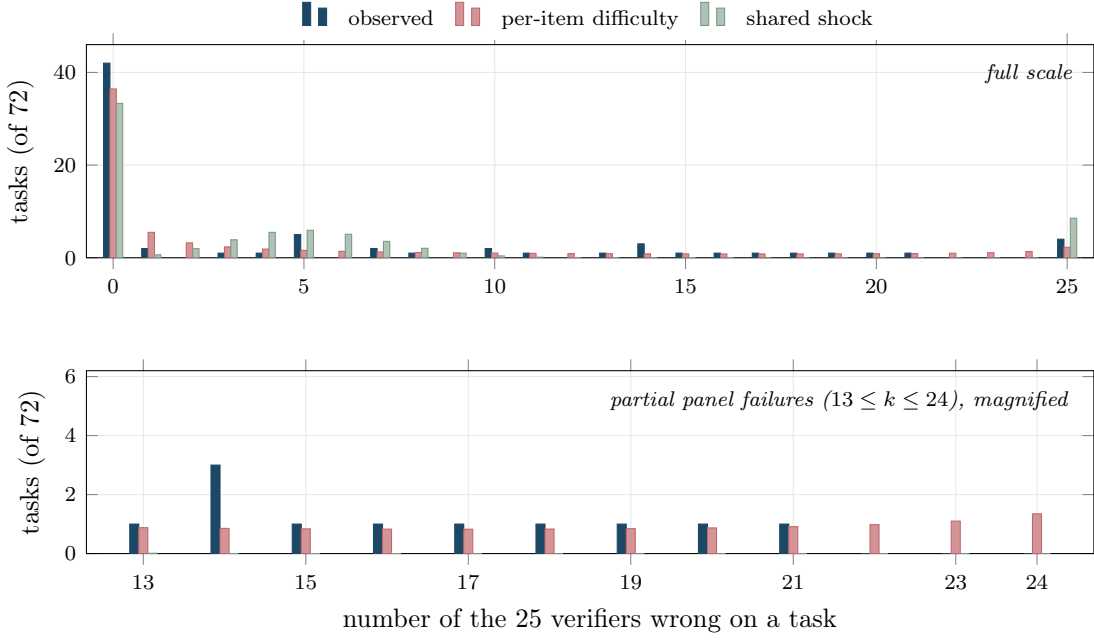


Figure 2: Where each model puts its mass, at full scale (top) and magnified over the partial-failure region (bottom). A 25-verifier majority fails when $k \geq 13$; it fails *partially* when $13 \leq k \leq 24$, a majority wrong while a minority is still right. That is how 11 of the panel’s 15 real failures happen, and shared shock is flat against zero across the whole region (0.01 tasks expected) — conditional on the shock it produces unanimity, conditional on no shock a tight binomial spike near $k = 5$ that dies well before 13. It compensates by over-predicting unanimous failure (8.5 against 4 observed). The per-item difficulty model reproduces the region at the same parameter count (11.1 predicted against 11 observed).

The fitted correlation lands on the independently measured ρ , the partial failure count is reproduced almost exactly, and the panel error is $3.7\times$ closer. Generalising over the 441-cell grid, the per-item model predicts more error than shared shock in 438/441 cells, by a median factor of $1.27\times$ and up to $2.71\times$. **Recommendation: parameterise verifier dependence by an item-difficulty distribution, not by a shared-shock probability.** ρ remains a useful summary of a panel; it is not a sufficient statistic for its error.

6 The quorum frontier and the irreducible blind spot

Majority vote is one point on a curve. Sweeping the acceptance threshold from 1 to 25 votes gives the panel’s full frontier (Figure 3).

Before reading anything off that curve, it must clear the screen §3.1 demanded of the LLM panel: a verification panel must beat a rule that ignores the code. On this corpus “reject everything unread” errs on the 26 viable candidates, a rate of 0.3611. **The panel does not beat that baseline until $need = 6$, and at $need = 5$ it exactly ties it.** Majority (0.2083) and the whole high-quorum region clear it comfortably, but the left third of the frontier does not, and nothing should be concluded from that region. We apply the test to ourselves because §3.1 applies it to others.

At unanimity the three numbers separate cleanly, and neither two-parameter model lands on the truth:

$$\text{real} = 0.0556, \quad \text{shared shock} = 0.1186 \text{ (} 2.13 \times \text{ high)}, \quad \text{per-item} = 0.0313 \text{ (} 1.77 \times \text{ low)}.$$

The disagreement is structural rather than a fitting artifact.

- **Shared shock** puts $P(\text{all } n \text{ wrong}) = \rho\mu + (1 - \rho)\mu^n$, tending to a hard floor $\rho\mu = 0.1186$

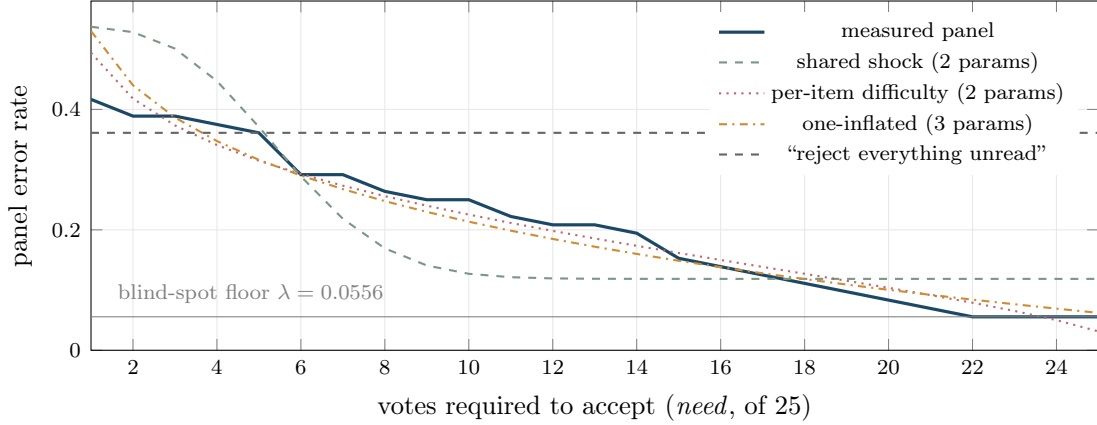


Figure 3: The real quorum frontier against three models, with the trivial baseline. RMSE over all 25 thresholds: shared shock 0.0766, per-item difficulty 0.0251, one-inflated 0.0328. From $need = 17$ upward shared shock is frozen at 0.1186 while reality keeps falling; past $need = 22$ reality stops at 0.0556 while the per-item model keeps promising improvement. Note the left of the plot: the panel does not beat “reject everything unread” until $need = 6$, and nothing should be concluded from that region.

at *every* panel size. The model asserts that past some threshold, raising the quorum cannot help at all.

- **Per-item difficulty** puts $P(\text{all } n \text{ wrong}) = \mathbb{E}[d^n] = B(\alpha + n, \beta)/B(\alpha, \beta)$, which decays as $n^{-\beta}$ with $\beta = 0.5762$ — slowly, but to zero. The model asserts there is no floor, and that enough verifiers will eventually verify anything.
- **Reality has a floor, but not shared shock’s.** It sits at $\lambda = 0.0556$: four defects that *every one* of the 25 verifiers misses. That is a shared blind spot of the panel, not a correlated shock. It is also the region Eckhardt and Lee [5] identified as the binding constraint — intensity mass at $\theta \geq 0.5$, which caps achievable reliability “regardless of the magnitude of the average component failure probability”.

The distinction is operationally important. A shared shock is a property of the *correlation*; a blind spot is a property of *which verifiers you chose*. You escape the first by decorrelating an existing panel, and the second only by adding a verifier of a different kind. Adding one-inflation — a λ -atom on “all wrong” plus a beta-binomial over the rest — captures both, at the cost of a third parameter. Once the blind spot is represented explicitly, the correlation remaining in the reducible units is 0.4513 rather than the marginal 0.5873; feeding the marginal value *and* arming the atom would double-count the same shared failures.

Finally, the assumed shape — not the correlation — decides the aggregation rule. Sweeping the cost-optimal threshold q^* over correlation, base rate and false-accept/false-reject cost ratio, at $n = 25$ the optimum spans 12–14 under shared shock but 1–25 under per-item difficulty, with shifts as large as ± 11 votes in the same cell. Under shared shock, tuning the quorum on this panel appears to buy nothing (1.00 $\times$); on the real panel it is the single largest available error reduction (3.75 $\times$, from 0.2083 at majority to 0.0556).

7 Consequences for panel design

7.1 Group balancing is not a safe default

If declared groups are the problem, weighting by them looks like the fix: give each declared independence group one vote instead of counting every verifier equally. Under shared shock this crosses over from harmful to helpful at $\rho = 0.25$ for a reference $[7, 1, 1, 1, 1]$ panel at $p = 0.75$, and

that crossover survives the change of shape — it sits at $\rho = 0.25$ under both models, a robustness result.

It does not survive the groups being fake. When task difficulty is shared across groups rather than within them — the regime the measured panel’s 0.526 cross-group correlation points toward — group balancing **never wins at any correlation**, and loses by up to 3 percentage points:

ρ	naive majority	group-balanced	winner
0.000	0.0345	0.0659	naive
0.125	0.1200	0.1366	naive
0.250	0.1719	0.1809	naive
0.500	0.2232	0.2255	naive
0.750	0.2448	0.2449	naive
1.000	0.2492	0.2492	tie

The mechanism is straightforward: balancing pays a real cost, discarding the extra votes inside the large group, and that cost is only worth paying when the small groups supply evidence the large group lacks. When every group errs on the same hard items, the discarded votes are pure loss. At $\rho = 1$ both rules report $\approx 1 - p$: the panel has collapsed to a single verifier and no aggregation rule can recover anything.

7.2 Correlation, not accuracy, is the exploitable surface

A last consequence concerns adversarial pressure, and we state its status plainly: this is a simulation result built on the measured panel’s parameters, not a second measurement. In a matched-budget search benchmark, contributors optimise to pass the gate — drawing k candidates and submitting the one that looks best on exactly the quantity the gate ranks. Holding per-verifier accuracy fixed at the measured $p = 0.7956$ and the head count fixed at 25, we vary only the *dependence structure*. Because a panel’s dependence has two components that our own §6 insists are different mechanisms, we run each alone rather than moving both together:

panel ($n = 25, p = 0.7956$)	correlation	blind spot	catastrophic seeds
independent	0.0	0.0	0
blindspot_only	0.0	0.0556	14
correlated_only	0.4513	0.0	53
measured	0.4513	0.0556	58

Catastrophic seeds are out of 100, for the best-performing search strategy, at the highest adversary fraction (0.2) and effort ($k = 8$). The two end rows reproduce the published matrix exactly; the two middle rows are the single-factor cells this paper adds.

Correlation carries the effect. Correlation alone reproduces 53 of the 58 catastrophic seeds — 91% — while the blind spot alone reproduces 14. The same head count and the same per-verifier accuracy separate immunity from near-coin-flip failure, and it is the correlation component that does it. Two honest caveats: the components are *sub-additive* ($53 + 14 = 67$ against a joint 58), so they overlap rather than compose, and a blind spot of 0.0556 is a much smaller perturbation than a correlation of 0.4513, so this compares the two effects *as measured on our panel* rather than at matched intensity. What the decomposition rules out is the alternative reading — that the gap is really the blind spot, with correlation along for the ride.

A second observation from the same benchmark is worth recording because it is a measurement trap. A panel that errs at the same rate for everyone is not a panel that harms everyone equally: an error only hurts in the direction a producer is exposed to. Strategies whose proposals are almost always viable (base viability ≥ 0.89) can essentially only suffer false *rejects*; a strategy proposing viable work 40% of the time is mainly exposed to false *accepts*. Across five strategies spanning a

base-viability range of 0.588, the least viable one took nine times the utility damage of the next worst. Its *accept rate went up* as the gate degraded, from 0.398 to 0.488, while every healthy strategy’s fell. **Reading accept rate as gate health inverts the ranking: the strategy whose throughput improved was the one being destroyed.**

8 Threats to validity

The panel’s diversity is constructed. This is the primary threat. Our 25 verifiers are 5 input regions $\times$ 5 seeds, a diversity structure we designed so that the declared independence label would be testable. It is not a sample of the diversity structures found in deployed review panels, and the measured $\rho = 0.5873$ should be read as one real value, not as an estimate of a population parameter. The qualitative findings — that a declared label under-predicts dependence, that failures are partial, that a blind spot exists — are the transferable claims; the specific numbers are not.

The per-item difficulty model is validated against one panel. It fits far better than shared shock, which is a low bar. A Beta difficulty distribution is a modelling choice — the natural two-parameter conjugate form — and not a shape we measured. Both models also treat verifiers as exchangeable, with identical accuracy and one common correlation; real panels are heterogeneous, and while we showed heterogeneity does not by itself explain the shared-shock gap, neither model represents it.

Corpus scale and domain. 72 candidates from 24 problems, all self-contained algorithmic programs with executable hidden tests. The blind-spot floor rests on four candidates. The corpus supports a base rate (0.639 non-viable) that would differ in a repository where most submitted work is roughly correct, and §6’s cost-optimal thresholds are explicitly base-rate dependent.

The LLM result is about small models. §3.1’s panel used four open-weight models in the 1–2B parameter range. It establishes that verifier competence must be screened before dependence is estimated; it does not establish that larger models fail the screen, and we expect they do not.

The adversarial results are simulated. §7.2’s decomposition is generated by a search benchmark whose verification panel is parameterised from our measurements, not observed in a live system. The single-factor cells separate correlation from blind spot *within that simulator*; they do not establish which component an attacker on a real review gate would exploit. In that simulator the panel reads each candidate through a single viability bit, so questions of the form “can an attacker learn what the panel cannot see” are unanswerable within the instrument: the blind spot has no address in artifact space. We report the contrast as a consequence of the measured parameters, not as evidence about real attackers.

Quorum restriction in the grid. The 441-cell comparison of §4.3 is restricted to $q = 0.5$, where “the panel is wrong” and “the panel wrongly accepts” coincide. Above 0.5 they diverge, and separating false accepts from false rejects requires the corpus base rate — which is why the quorum analysis of §6 is done directly on the measured panel rather than on the grid.

9 Reproduction

All artifacts are committed in the IDKMesh repository [16]. The panel votes, the corpus with its execution-decided labels, and the simulation grid are under `experiments/results/`; the analyses are pure standard-library Python.

```
python3 sim/e017_verify.py --seeds 5 --out e017-votes.jsonl # regenerate votes
python3 sim/e017_analyze.py experiments/results/E017-partial-oracle-votes.jsonl.gz
python3 sim/e018_dependence_models.py # the 441-cell model comparison
python3 sim/e020_quorum_frontier.py # the quorum frontier, deterministic
python3 paper/make_figures.py # regenerate every figure in this paper
PYTHONPATH=. python3 paper/deconfound_panel.py \
--seeds 100 --agents 64 --generations 50 --change-at 25 --jobs 4
```

The figures are generated by importing the same analysis modules the experiment records use, so no number in this paper can drift from the artifact it came from. The last command runs §7.2’s single-factor cells; it registers the two extra panels on the benchmark’s own panel table at run time, so it adds cells without modifying any committed experiment module. Full experiment records, including the preregistered predictions that were falsified, are in `experiments/E015–E020`.

10 Conclusion

A verification panel’s head count is a budget, not a measurement. On a panel where every member demonstrably works and every error is a genuine missed defect, 25 verifiers delivered less evidence than the best one of them, the standard independence discount overstated what was available *on this panel*, and the dependence model in common use was wrong in shape rather than in parameter — unable to produce the partial failures that are how real panels actually fail. Beneath all of that sat a floor no aggregation rule could reach: defects invisible to every verifier in the panel, escapable only by adding a verifier of a different kind.

The operational summary is short. Screen your verifiers for discrimination before you trust any statistic about them. Estimate independence from observed error vectors, never from declared diversity. Establish the dependence *shape* before applying $N/(1 + (N - 1)\rho)$, which overstates independence when difficulty is shared across items. Check your panel against a rule that ignores the artifact. Measure your quorum, since it is likely a larger lever than your panel size. And treat the dependence structure, not accuracy, as the property an adversary will find. Every one of these is a measurement instruction, which is the paper’s real conclusion: a panel’s evidentiary value is not a property you can assume from its size, its members’ accuracy, or its declared diversity.

Data availability. Code, artifacts, and experiment records: <https://github.com/MSKazemi/idxmesh>.

AI/tool provenance. Drafting and revision of this manuscript, and the analysis and figure-generation code in §9, were materially assisted by an AI coding assistant. Every quantitative claim reported here was independently re-derived from the committed artifacts by re-running the cited scripts rather than accepted from prose, and the manuscript was revised against a prior adversarial review (§9’s repository carries that review alongside this source). AI assistance is not represented as independent human or external scientific review, and this manuscript has not undergone peer review at a venue as of this revision.

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